



Web3 & Modular Blockchain

How Builders Are Rethinking Infrastructure

By Samuel Je Christian · Web3 Corner Date: May 17, 2026



One Blockchain to Rule Them All — Or Not

For years, the Web3 conversation went like this: *"Which blockchain will win?"* Ethereum vs. Solana. Bitcoin vs. Ethereum. Every new chain claiming to be faster, cheaper, and more decentralized.

The assumption behind these debates was simple: one blockchain would eventually do everything — store data, execute code, reach consensus, and settle transactions — all in one place.

That assumption is being dismantled in 2026.

The new reality: builders don't choose one blockchain. They assemble a custom stack of specialized components — each optimized for one specific job.

This is **modular blockchain architecture**. And it's quietly reshaping how Web3 is built.



The Problem With Monolithic Blockchains

A monolithic blockchain does everything in one place:

-  **Consensus:** Orders and validates transactions.
-  **Execution:** Runs smart contract code.
-  **Data Availability (DA):** Stores and publishes transaction data.
-  **Settlement:** Settles disputes and finalizes state.

The problem: optimizing for all four simultaneously creates impossible trade-offs.

- **Ethereum:** Highly secure and decentralized, but slow and expensive at peak demand.
- **Solana:** Extremely fast and cheap, but sacrifices some decentralization through higher hardware requirements.

[!IMPORTANT] Every monolithic chain faces the same **scalability trilemma**: you can have two of three — security, scalability, decentralization — but never all three at once. Modular architecture solves this by separating the four functions and letting each layer specialize.



The Four Layers of a Modular Stack

1. Execution Layer

Where smart contracts run and user transactions are processed.

- **Examples:** Ethereum L2s (Arbitrum, Optimism, Base), StarkNet, zkSync, and Solana (as a fast execution layer).
- **Innovation: Arbitrum Stylus** — now supports Rust and C++ alongside Solidity.
- **Focus:** Competes purely on speed, throughput, and developer experience.

2. Data Availability (DA) Layer

Where transaction data is published so anyone can verify the blockchain's state.

This is the most critical — and often most overlooked — layer in Web3. If data isn't available, fraud proofs can't work. If you can't verify the data, you can't trust the chain.

- **Examples:** Ethereum (still the most trusted DA layer), Celestia (modular DA pioneer), EigenDA (built on EigenLayer), and Avail (Polygon's DA layer).
- **2026 Insight:** Execution and data availability have emerged as the two business lines that define token demand and profitability in the modular blockchain cycle.

3. Consensus Layer

Where the network agrees on the order and validity of transactions.

- **Example:** Ethereum's Beacon Chain handles consensus separately from execution — a key innovation of The Merge in 2022.

4. Settlement Layer

Where final dispute resolution happens. The ultimate source of truth.

- **Example:** Ethereum L1 serves as the settlement layer for most of the modular ecosystem — the bedrock everything else builds on.

Why Builders Are Going Modular in 2026

In 2026, development teams abandon the monolith logic. They no longer pick a single blockchain and accept all the design trade-offs that choice imposes. Now they assemble their own stack with independent modular components.

-  **A gaming app needs:** Fast execution (Arbitrum or Base) + Cheap data availability (Celestia) + Secure settlement (Ethereum L1).
-  **A DeFi protocol needs:** High-security execution (zkSync or StarkNet) + Ethereum-grade data availability + Ethereum settlement for maximum trust.
-  **An AI x blockchain system needs:** Specialized compute execution (custom) + Verifiable data availability + Cross-chain oracle integration (Chainlink).

The modular stack lets builders mix and match — choosing the best tool for each job instead of accepting a one-size-fits-all chain.

Real Modular Projects in 2026

-  **Celestia:** The pioneer of modular data availability. It decouples DA from execution entirely, letting any chain publish data to Celestia instead of using expensive Ethereum DA.
-  **EigenLayer:** A restaking protocol that lets Ethereum validators secure additional networks. It creates a modular security marketplace, allowing new chains to borrow Ethereum's security without building from scratch.
-  **Arbitrum Stylus:** Syndicate integrated Stylus, the execution environment of Arbitrum, into the architecture of SyndDB — its open-source SQLite database replication system —

reducing TEE attestation verification time significantly. Developers can now write smart contracts in Rust and C++ alongside Solidity.

-  **Polygon AggLayer:** Connects multiple chains into one unified liquidity and state layer for modular interoperability at scale.
-  **Avail:** Polygon's dedicated DA layer providing cheap, fast data availability for any chain that needs it.

The Honest Trade-Offs

- **Complexity:** Modular stacks are significantly more complex to build and audit than monolithic chains. More moving parts equal more potential failure points.
- **Fragmentation:** Different execution and DA layers can fragment assets and liquidity. Cross-layer bridging introduces risk (e.g., the ThorChain \$10.8M exploit this week).
- **Immaturity:** Many modular components (Celestia, EigenDA, Avail) are still early and are not yet fully battle-tested at multi-year scale.
- **Coordination Risk:** When an issue arises, determining which layer is responsible (the builder, the DA layer, or the settlement chain) gets complex fast.

The One-Line Summary

Modular blockchain architecture separates execution, data availability, consensus, and settlement into specialized layers — letting builders assemble custom stacks optimized for their needs instead of accepting the trade-offs of one monolithic chain.

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